

Documentation of the model implemented in Python

Comparison with the paper formulation

1 Introduction

This document describes the MIQP model implemented in the file `model.py` and highlights the main differences with respect to the mathematical formulation in the paper *Drone Coverage Path Planning* (Amorosi, Dell’Olmo, Puerto, Valverde).

Common notation

- $\mathcal{R}$: set of regions.
- $\mathcal{V}'$: vertices (interruption points in each region).
- $\mathcal{E}^{int} = \mathcal{E}_{RL}^{int} \cup \mathcal{E}_{LR}^{int}$: intra-region edges (RL = forward, LR = backward).
- $\mathcal{E}^{ext}$: inter-region edges.
- $\mathcal{T}_r$: polygonal chains available for region r .
- $\mathcal{S}_t$: segments of chain t .
- Q^s, Q^{s+1} : endpoints of segment s .
- $L(s) = \|Q^{s+1} - Q^s\|$: length of segment s .
- $E^{xy} = \frac{1}{2}cA$, $E^z = \frac{1}{2}cA$: drag constants.
- ρ_t : air density at the height of chain t .
- $\nu_{t,s}^d$: actual drone speed over segment s of chain t .
- W^{max} : maximum drone endurance.

2 Variables of the implemented model

2.1 Binary variables

Variable	Domain	Description
μ_v^{ts}	$\mathcal{V}' \times \{(t, s) : t \in \mathcal{T}_r, s \in \mathcal{S}_t\}$	1 if vertex $v = (r, i)$ is assigned to segment s of chain t .
α_t	$\{t \in \mathcal{T}_r : r \in \mathcal{R}\}$	1 if chain t is selected for its region.
x_{uv}^o	$\mathcal{E} \times \mathcal{O}$	1 if the drone traverses (u, v) in operation o .
y_v^o	$\mathcal{V} \times \mathcal{O}$	1 if vertex v is visited in operation o .
ζ^o	$\mathcal{O}$	1 if all vertices have been visited before operation o .
$\varphi_{rr'}$	$\mathcal{R} \times \mathcal{R}, r \neq r'$	1 if the drone moves from r to r' at the height of r .
$\psi_{rr'}$	$\mathcal{R} \times \mathcal{R}, r \neq r'$	1 if the height of r is lower than that of r' .

2.2 Continuous and integer variables

Variable	Range	Description
P_v	$\mathbb{R}^3$	Coordinates of the point associated with vertex v .
γ_v	$[0, 1]$	Position within the selected segment.
λ_v	$[0, S_t]$	Global parametrization on the selected chain.
dfs_v	$[0, L_{max}]$	Distance from the start of the chain.
$efsv$	$[0, L_{max} \cdot 10]$	Energy from the start of the chain.
d_{uv}^{xy}	$[0, D_{max}]$	Inter-region horizontal distance.
d_{uv}^z	$[0, H_{max}]$	Inter-region vertical distance.
d_{uv}	$[0, D_{max} + H_{max} + L_{max}]$	Total distance between u and v .
w_{uv}	$[0, \infty)$	Energy between u and v .
k^o	$\{0, \dots, \mathcal{V}' \}$	Number of vertices visited in operation o .
u_v^o	$\{1, \dots, \mathcal{V}' \}$	MTZ variable for subtour elimination.
ρ_r	$[0, \rho(0)]$	Selected density for region r .
η_v^{ts}	$[0, 1]$	Linearization of $\gamma_v \cdot \mu_v^{ts}$.

3 Location constraints

They match the paper in structure. The main difference is in the fixing of the START and END vertices:

$$P_v = \sum_{t \in \mathcal{T}_r} \sum_{s \in \mathcal{S}_t} [\mu_v^{ts} Q^s + \eta_v^{ts} (Q^{s+1} - Q^s)], \quad \forall v = (r, i) \in \mathcal{V}', \quad (\text{LC1})$$

$$P_v^z = \sum_{t \in \mathcal{T}_r} \sum_{s \in \mathcal{S}_t} \mu_v^{ts} \cdot (\text{height of } t), \quad \forall v = (r, i) \in \mathcal{V}', \quad (\text{LC1z})$$

$$\sum_{s \in \mathcal{S}_t} \mu_v^{ts} = \alpha_t, \quad \forall v = (r, i) \in \mathcal{V}', \quad \forall t \in \mathcal{T}_r, \quad (\text{LC2})$$

$$\sum_{t \in \mathcal{T}_r} \alpha_t = 1, \quad \forall r \in \mathcal{R}, \quad (\text{LC3})$$

$$\lambda_v = \sum_{t \in \mathcal{T}_r} \sum_{s \in \mathcal{S}_t} (s-1) \mu_v^{ts} + \gamma_v, \quad \forall v \in \mathcal{V}', \quad (\text{LC4})$$

$$\lambda_v = 0, \quad \forall v = (r, 1), (r, 2), \quad (\text{LC5})$$

$$\mu_v^{t0} = \alpha_t, \quad \forall v = (r, 1), (r, 2), \quad \forall t \in \mathcal{T}_r, \quad (\text{LC5b (new)})$$

$$\lambda_u \leq \lambda_v, \quad \forall (u, v) \in \mathcal{E}_{RL}^{int}, \quad (\text{LC6})$$

$$\lambda_u = \lambda_v, \quad \forall (u, v) \in \mathcal{E}_{LR}^{int}, \quad (\text{LC7})$$

$$\lambda_v = \sum_{t \in \mathcal{T}_r} S_t \alpha_t, \quad \forall v = (r, 2R+1), (r, 2R+2), \quad (\text{LC8})$$

$$\mu_v^{t, S_t-1} = \alpha_t, \quad \forall v = (r, 2R+1), (r, 2R+2), \quad \forall t \in \mathcal{T}_r. \quad (\text{LC8b (new)})$$

Constraints LC5b and LC8b explicitly fix the segment (first or last) of the START and END vertices, a simplification with respect to the paper where this fixing was done implicitly through λ_v and the domain of μ .

4 Drone path constraints

They match the paper except for subtour elimination:

$$\sum_{v \neq v_0} x_{v_0 v}^o = 1 - \zeta^o, \quad \forall o \in \mathcal{O}, \quad (\text{DP1})$$

$$\sum_u x_{uv}^o - \sum_u x_{vu}^o = 0, \quad \forall v \in \mathcal{V}', \quad \forall o \in \mathcal{O}, \quad (\text{DP2})$$

$$\sum_{v \neq v_0} x_{vv_0}^o = 1 - \zeta^o, \quad \forall o \in \mathcal{O}, \quad (\text{DP3})$$

$$\sum_u x_{vu}^o = y_v^o, \quad \forall v \in \mathcal{V}, \forall o \in \mathcal{O}, \quad (\text{DP4})$$

$$\sum_u x_{uv}^o = y_v^o, \quad \forall v \in \mathcal{V}, \forall o \in \mathcal{O}, \quad (\text{DP5})$$

$$\sum_o y_v^o = 1, \quad \forall v \in \mathcal{V}', \quad (\text{DP6})$$

$$u_u^o - u_v^o + 1 \leq (|\mathcal{V}'| - 1)(1 - x_{uv}^o), \quad \forall u \neq v \in \mathcal{V}', \forall o \in \mathcal{O}, \quad (\text{DP7 (MTZ)})$$

$$\sum_o x_{uv}^o = 1, \quad \forall (u, v) \in \mathcal{E}_{RL}^{int}. \quad (\text{DP8})$$

The paper's main formulation uses the same MTZ constraints. The paper also describes an alternative edge-based formulation that replaces MTZ with Dantzig–Fulkerson–Johnson (DFJ) exponential subtour elimination constraints, separated dynamically via a lazy constraint callback during the branch-and-cut search (see Section 11 below). Both are implemented in the codebase ¹.

5 Intra-region distances

Paper approach (with bilinear β)

The paper defines $d_{vv'}^{ss'}$ using bilinear products $\beta_{vv'}^{ss'} = \mu_v^s \mu_{v'}^{s'}$, resulting in expressions depending on 4 indices (two vertices, two segments).

Implemented approach (DFS)

Instead of β , a continuous variable dfs_v is introduced to represent the cumulative distance from the start of the chain to vertex v :

$$dfs_v = \sum_{t \in \mathcal{T}_r} \sum_{s \in \mathcal{S}_t} [\text{cum_before}(s) \cdot \mu_v^{ts} + L(s) \cdot \eta_v^{ts}], \quad \forall v \in \mathcal{V}', \quad (\text{DFS})$$

$$d_{uv} = dfs_v - dfs_u, \quad \forall (u, v) \in \mathcal{E}_{RL}^{int}, \quad (\text{IDIST})$$

$$d_{uv} = 0, \quad \forall (u, v) \in \mathcal{E}_{LR}^{int}. \quad (\text{IDIST0})$$

¹The vertex-based model is `RingsModel` in `mip_rings.py`, the edge-based model is `EdgesModel` in `mip_edges.py`.

where $\text{cum_before}(s) = \sum_{s' < s} L(s')$, and $\eta_v^{ts} = \gamma_v \cdot \mu_v^{ts}$ is linearized via standard Fortet constraints (binary $\times$ continuous).

This formulation is more compact: it avoids the double products $\mu_v^s \mu_{v'}^{s'}$ and reduces the number of variables and quadratic constraints.

6 Intra-region energy

Analogously to distance, a variable efs_v is used for cumulative energy:

$$efs_v = \sum_{t \in \mathcal{T}_r} \sum_{s \in \mathcal{S}_t} [C_{ts} \cdot \text{cum_before}(s) \cdot \mu_v^{ts} + C_{ts} \cdot \eta_v^{ts}], \quad \forall v \in \mathcal{V}', \quad (\text{EFS})$$

$$w_{uv} = efs_v - efs_u, \quad \forall (u, v) \in \mathcal{E}_{RL}^{int}, \quad (\text{IENER})$$

$$w_{uv} = 0, \quad \forall (u, v) \in \mathcal{E}_{LR}^{int}, \quad (\text{IENER0})$$

where $C_{ts} = E^{xy} \rho_t \nu_{ts}^d L(s)$ is the energy per unit distance on segment s of chain t , computed as a constant (pre-process).

7 Inter-region distances and energy

Distances

They match the paper:

$$d_{uv} = d_{uv}^{xy} + d_{uv}^z, \quad \forall (u, v) \in \mathcal{E}^{ext}, \quad (\text{DIST3})$$

$$d_{uv}^{xy} \geq \|(P_u^x - P_v^x, P_u^y - P_v^y)\|_2, \quad \forall (u, v) \in \mathcal{E}^{ext}, \quad (\text{DIST4 (SOC)})$$

$$d_{uv}^z \geq |P_u^z - P_v^z|, \quad \forall (u, v) \in \mathcal{E}^{ext}. \quad (\text{DIST5-6})$$

Inter-region energy

The inter-region energy in the implementation follows the same structure as the paper (with $\varphi_{rr'}$ and $\psi_{rr'}$), but all bilinear products are explicitly linearized using the functions `_lin_bb` (binary $\times$ binary) and `_lin_bc` (binary $\times$ continuous).

The general implemented form is:

$$w_{uv} = \underbrace{B_{r1}^{xy} \varphi_{rr'} d_{uv}^{xy} \sum_{t \in \mathcal{T}_r} \alpha_t \rho_t}_{\text{linearized}} + \underbrace{B^z \varphi_{rr'} \psi_{rr'} d_{uv}^z \frac{1}{2} (\rho_r + \rho_{r'})}_{\text{linearized}}$$

$$\begin{aligned}
& \underbrace{+ B_{r2}^{xy}(1 - \varphi_{rr'})d_{uv}^{xy} \sum_{t \in \mathcal{T}_{r'}} \alpha_t \rho_t}_{\text{linearized}} + \underbrace{B^z(1 - \varphi_{rr'})(1 - \psi_{rr'})d_{uv}^z \frac{1}{2}(\rho_r + \rho_{r'})}_{\text{linearized}}, \\
& \text{(INTER-ENER)}
\end{aligned}$$

where each bilinear term is replaced by an auxiliary variable with its 4 linearization constraints.

8 Depot energy

The implementation uses a simplified model with constant density ρ_0 :

$$\begin{aligned}
d_{v_0v} &= d_v^{xy} + d_v^z, & \forall v \in \mathcal{V}', & \text{(DEP-DIST)} \\
d_v^{xy} &\geq \|(P_v^x - d_x, P_v^y - d_y)\|_2, & \forall v \in \mathcal{V}', & \text{(DEP-SOC)} \\
d_v^z &\geq |P_v^z|, & \forall v \in \mathcal{V}', & \text{(DEP-Z)} \\
w_{v_0v} &= E^{xy} \rho_0 \nu^{xy} d^{xy} + \frac{1}{2} E^z \rho_0 \nu^z d^z, & \forall v \in \mathcal{V}', & \text{(DEP-ENER)} \\
w_{vv_0} &= E^{xy} \rho_0 \nu^{xy} d^{xy}, & \forall v \in \mathcal{V}', & \text{(DEP-ENER-S)}
\end{aligned}$$

Difference: In the paper, the ascent energy ($v_0 \rightarrow v$) considers the variable density $\rho_r + \rho_0$; the implementation uses only ρ_0 as an approximation.

9 Endurance constraint

$$\begin{aligned}
\sum_{u,v \in \mathcal{V}} \ell_{uv}^o &\leq W^{max}, & \forall o \in \mathcal{O}, & \text{(END)} \\
\ell_{uv}^o &\leq x_{uv}^o \cdot M, & \forall (u,v) \in \mathcal{E}, \forall o \in \mathcal{O}, & \text{(END-LIN1)} \\
\ell_{uv}^o &\leq w_{uv}, & \forall (u,v) \in \mathcal{E}, \forall o \in \mathcal{O}, & \text{(END-LIN2)} \\
\ell_{uv}^o &\geq w_{uv} - (1 - x_{uv}^o)M, & \forall (u,v) \in \mathcal{E}, \forall o \in \mathcal{O}, & \text{(END-LIN3)} \\
\ell_{uv}^o &\geq 0, & \forall (u,v) \in \mathcal{E}, \forall o \in \mathcal{O}. & \text{(END-LIN4)}
\end{aligned}$$

where ℓ_{uv}^o linearizes the product $x_{uv}^o \cdot w_{uv}$.

10 Objective function

$$\min W = \sum_{u,v \in \mathcal{V}} \sum_{o \in \mathcal{O}} \ell_{uv}^o, \quad \ell_{uv}^o \text{ linearizes } x_{uv}^o \cdot d_{uv}. \quad \text{(OBI)}$$

Difference: The paper minimizes total distance; the implementation does exactly the same, but with explicit linearization of the product $x \cdot d$.

11 Edge-based alternative (EdgesModel)

Alongside `RingsModel`, the codebase implements an alternative formulation in `mip_edges.py` (class `EdgesModel`) that replaces vertex-visitation variables y_v^o with edge-coverage variables y_{uv}^o defined on the set of intra-ring edges $\mathcal{E}^{int}$, following a perspective akin to the rural postman problem.

Variables

The binary variables y_{uv}^o indicate whether the intra-ring edge (u, v) is covered in operation o . The integer variable k^o counts the number of covered edges instead of the number of visited vertices. All other variables ($x_{vv'}^o$, ζ^o , α_t , μ_v^{ts} , γ_v , λ_v , P_v , etc.) remain unchanged. The MTZ auxiliary variables u_v^o are omitted.

Modified constraints

The flow conservation (DP2) and depot constraints (DP1, DP3) are kept. The degree constraints DP4–DP5 are relaxed to ≤ 1 (at most one edge leaves or enters each vertex per operation). The vertex-visitation equalities DP6 are removed, replaced by edge-coverage linking constraints:

$$\sum_{o \in \mathcal{O}} y_{uv}^o = 1, \quad \forall (u, v) \in \mathcal{E}^{int}, \quad (\text{EC1})$$

$$y_{uv}^o = x_{uv}^o, \quad \forall (u, v) \in \mathcal{E}^{int}, \forall o \in \mathcal{O}. \quad (\text{EC2})$$

Subtour elimination: DFJ lazy cuts

The MTZ subtour elimination is not used. Instead, the model enforces subtour elimination via the exponential family of Dantzig–Fulkerson–Johnson (DFJ) constraints, separated dynamically through a lazy constraint callback:

$$\sum_{u \in S, v \notin S} x_{uv}^o \geq 1, \quad \forall S \subsetneq \mathcal{V} : v_0 \notin S, \forall o \in \mathcal{O}, \quad (\text{DFJ})$$

At each feasible integer incumbent, a breadth-first search from the depot identifies, for each operation o , the connected components of the subgraph induced by edges with $x_{uv}^o = 1$. Any component that does not contain the depot triggers the addition of cut (DFJ). The callback is implemented using Gurobi’s MIPSOL callback with `LazyConstraints=1`.

Idle operations

An additional constraint forces all traversal variables to zero in idle operations ($\zeta^o = 1$), preventing the lazy callback from adding spurious cuts:

$$\sum_{u,v} x_{uv}^o \leq |\mathcal{V}|^2 (1 - \zeta^o), \quad \forall o \in \mathcal{O}. \quad (\text{IDLE})$$

Relationship to the paper

The alternative edge-based formulation described in Section 4.2 of the paper matches this implementation exactly: the paper’s relaxed degree constraints ($D'_{\text{path-C}_4}$, $D'_{\text{path-C}_5}$) correspond to the ≤ 1 degree bounds above; the DFJ lazy cut separation is described in the paper under “Drone path constraints (edge-based)”; and the edge-coverage linking (EC_1 , EC_{2a} , EC_3) is identical.

12 Summary of differences

Aspect	Paper	Implementation
Intra-RL distance	$\beta_{vv'}^{ss'} = \mu_v^s \mu_{v'}^{s'}$ (bilinear)	dfs_v with $\eta_v^{ts} = \mu_v^{ts} \gamma_v$ linearized
Intra-RL energy	Analogous to distance	Same efs_v strategy
Subtours (vertex-based)	MTZ (u_v^o)	MTZ (u_v^o)
Subtours (edge-based)	DFJ (lazy cuts)	DFJ (lazy cuts)
START/END fixing	Implicit (λ)	Explicit (LC5b, LC8b)
Depot energy $\uparrow$	$\rho_r + \rho_0$ variable	ρ_0 constant
Linearization	β bilinear not linearized	All bilinearities explicitly linearized
Number of heights	Multiple heights with φ, ψ	Multiple heights with φ, ψ

The main contribution of the implementation is the elimination of bilinear products $\mu_v^s \mu_{v'}^{s'}$ through the dfs_v -based formulation, which avoids $O(|\mathcal{V}'|^2 |\mathcal{S}_t|^2)$ variables β .